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One-class classification-based control charts for multivariate process monitoring

THUNTEE SUKCHOTRAT¹, SEOUNG BUM KIM^{1,*} and FUGEE TSUNG³

¹*Department of Industrial and Manufacturing Systems Engineering, University of Texas at Arlington, Arlington, TX 76019, USA*

²*Division of Information Management Engineering, Korea University, Seoul, South Korea*

E-mail: sbkim1@korea.ac.kr

³*Department of Industrial Engineering and Logistics Management, Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong*

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One-class classification problems have attracted a great deal of attention from various disciplines. In the present study, attempts are made to extend the scope of application of the one-class classification technique to Statistical Process Control (SPC) problems. New multivariate control charts that apply the effectiveness of one-class classification to improvement of Phase I and Phase II analysis in SPC are proposed. These charts use a monitoring statistic to represent the degree of being an outlier as obtained through one-class classification. The control limits of the proposed charts are established based on the empirical level of significance on the percentile, estimated by the bootstrap method. A simulation study is conducted to illustrate the limitations of current one-class classification control charts and demonstrate the effectiveness of the proposed control charts.

Keywords: Data mining, Hotelling's T^2 , multivariate process, one-class classification method, statistical process control

1. Introduction

Statistical Process Control (SPC) is one of the widely used techniques for quality control. The basic objective of SPC is to quickly detect the occurrence of special cause variation, so that the process can be investigated and corrective action may be taken before quality deteriorates and defective units are produced (Stoumbos *et al.*, 2000). One important tool in SPC is the control chart, which is used to monitor the performance of a process over time to keep the process in an in-control state. In general, control chart problems in SPC can be divided into two phases (Woodall and Montgomery, 1999; Woodall, 2000). Phase I analysis tries to isolate the in-control (baseline) data from an unknown historical data set and establish the control limits for future monitoring (Zhang and Albin, 2007). Phase II analysis monitors the process using control charts derived from the “cleaned” in-control data set from the Phase I analysis. With a simple plot of the set of monitoring statistics derived from the original samples, the control chart can effectively determine whether or not a process is in an in-control state. Examples of monitoring statistics include the sample average and the sample range. In addition to the monitoring statistics, another important component of control charts is control

limits, which often are calculated based on the probabilistic distribution of the monitoring statistic.

Hotelling extended the univariate control chart to handle multivariate problems (Hotelling, 1947). Hotelling's T^2 chart (T^2 chart) is a multivariate control chart that can monitor a multivariate process efficiently. T^2 charts use the T^2 statistic computed from the following equation:

$$T^2 = (\mathbf{x} - \bar{\mathbf{x}})^T \mathbf{S}^{-1} (\mathbf{x} - \bar{\mathbf{x}}), \quad (1)$$

where $\bar{\mathbf{x}}$ and $\mathbf{S}$ are a sample mean vector and a sample covariance matrix determined from the in-control (Phase I) data. The T^2 statistic measures the distance between an observation and the scaled-mean estimated from the in-control data. Given that $\mathbf{x}$ follows a multivariate normal distribution, the T^2 statistic follows an F distribution (Mason and Young, 2002). In T^2 charts, the $100\alpha\%$ tail area of an F distribution is used as the control limit, where α is the user-specified level of significance. It is known that T^2 charts can effectively control Type I and Type II error rates when the underlying distribution of the process data is the multivariate normal distribution (Lowry and Montgomery, 1995). However, the distributional assumption of T^2 charts means that it is difficult to apply these charts to the non-normal data that is often found in many modern industries. A number of non-parametric control charts have been developed to address the limitation of the

*Corresponding author

distributional assumptions (Chakraborti *et al.*, 2001; Liu *et al.*, 2004; Bakir, 2006; Kim *et al.*, 2007; Qiu, 2008). However, no consensus exists about which of them best satisfies all the conditions encountered in modern process systems. A detailed review of non-parametric control charts is beyond the scope of this paper.

As the limitations of current SPC techniques become increasingly obvious in the face of ever more complex processes, data mining algorithms, because of their proven capabilities to effectively analyze and manage large amounts of data, have the potential to resolve the challenging problems in SPC. Despite the enormous popularity of data mining studies that has seen their application to a variety of areas, few efforts have been made to integrate data mining algorithms with SPC (Smith, 1994; Cook and Chiu, 1998; Chinnam, 2002; Hwang *et al.*, 2005; Hu *et al.*, 2007). In particular, one-class classification methods share a common goal with control charts because both methods assume that the in-control group (target group) is the only population and it can be used for measuring the degree of abnormality of new observations. Several studies have been undertaken recently with the goal of implementing one-class classification algorithms as an alternative to traditional control charts. Sun and Tsung (2003) proposed kernel-distance-based charts (K charts) based on a Support Vector Data Description (SVDD) algorithm. SVDD is a modified version of the original Support Vector Machines (SVMs) concept for solving one-class classification problems. K charts use a monitoring statistic derived from the distance between the new observation and the decision boundary generated by the SVDD algorithm. The control limits of K charts are established and adjusted from a parameter in the SVDD algorithm. Sun and Tsung's study revealed that K charts perform better than T^2 charts when the data deviate from normality. Kumar *et al.* (2006) used another one-class SVM technique to construct robust K charts through normalized monitoring statistics. They showed that, in addition to the flexibility of non-normal data, robust K charts can efficiently handle autocorrelated process data. Furthermore, one-class SVM-based control charts have been applied to detect anomalies in computer-networking applications (Zhang *et al.*, 2007). It is clearly laudable that the aforementioned studies proposed to use the monitoring statistic from the one-class SVM method. Thus, the construction of the charts does not require any distribution assumptions. However, they did not suggest an efficient way to establish the control limits, one of the major components in control charts.

This paper makes contributions in two aspects. First, we propose an efficient way to establish the control limits necessary to improve the existing one-class SVM-based control charts. Second, we propose new one-class classification-based control charts based on a k -nearest-neighbor algorithm. Simulation studies were conducted to demonstrate the effectiveness of the proposed approaches in both the Phase I and Phase II analyses.

2. SVDD-based control charts

2.1. The SVDD algorithm

An SVM is one of the supervised learning algorithms popularly used for both regression and classification problems. SVMs use geometric properties and obtain a separating hyperplane by solving a convex optimization problem that simultaneously minimizes the generalization error and maximizes the geometric margin between the classes (Vapnik, 1998). Non-linear SVM models can be constructed from kernel functions such as linear, polynomial and radial basis functions, etc. SVDD is a mixture of SVM and the data description method for solving one-class classification problems (Tax and Duin, 2004). SVDD provides a hypersphere boundary around the data. A brief summary of the SVDD algorithm is as follows. Let $\mathbf{a}$ be the center of the hypersphere. Let R^2 be the radius of the hypersphere (i.e., the distance from $\mathbf{a}$ to the boundary). Let $\mathbf{x}_i = [x_{i1}, x_{i2}, \dots, x_{ip}]^T$, for $i = 1, 2, \dots, N$ be a sequence of p -variate training (target) observations. SVDD boundaries are constructed to minimize the volume of the hypersphere while maximizing the training observations captured by the hypersphere (Tax and Duin, 2004). That is, the problem is to

$$\min R^2 + C \sum_{i=1}^N \xi_i, \quad (2)$$

with the constraint:

$$\|\mathbf{x}_i - \mathbf{a}\|^2 \leq R^2 + \xi_i, \quad (3)$$

where $\xi_i > 0$ is the slack variable that allows $\mathbf{x}$ to be outside the hypersphere. C controls the trade-off between the volume of the hypersphere and the misclassification errors. Tax and Duin (2004) defined a user-specified parameter f that represents the fraction of the training data outside the decision boundary:

$$f = \frac{1}{NC}, \quad (4)$$

where N is the number of target observations. For instance, 80% of the training data points are supposed to be included in the SVDD boundary constructed with $f = 0.20$. When f is increased from 0.20 to 0.30, the volume of the hypersphere becomes smaller but the misclassification error in the target class becomes larger.

Equation (2) can be solved by the following Lagrangian:

$$L(R, \mathbf{a}, \alpha_i, \gamma_i, \xi_i) = R^2 + C \sum_{i=1}^N \xi_i - \sum_{i=1}^N \alpha_i \{R^2 + \xi_i - (\|\mathbf{x}_i - \mathbf{a}\|^2)\} - \sum_{i=1}^N \gamma_i \xi_i, \quad (5)$$

where $\alpha_i \geq 0$ and $\gamma_i \geq 0$ are the Lagrange multipliers. Setting partial derivatives of L with respect to R , $\mathbf{a}$ and ξ_i and

set to zero provides the following constraints:

$$\sum_{i=1}^N \alpha_i = 1, \quad (6)$$

$$\mathbf{a} = \sum_{i=1}^N \alpha_i \mathbf{x}_i, \quad (7)$$

$$\alpha_i = C - \gamma_i. \quad (8)$$

When substituting these constraints to Equation (5), the optimization problem becomes:

$$L = \sum_i \alpha_i (\mathbf{x}_i, \mathbf{x}_j) - \sum_{ij} \alpha_i \alpha_j (\mathbf{x}_i, \mathbf{x}_j). \quad (9)$$

The solution, the set of α_i , $i = 1, 2, \dots, N$, can be obtained by maximizing Equation (9) subject to $0 \leq \alpha_i \leq C$ and $\sum_{i=1}^N \alpha_i = 1$.

As with conventional SVM, the SVDD algorithm can generate more flexible decision boundaries by replacing the inner product with kernel functions. For example, the following Gaussian kernel function can be replaced with the inner product in Equation (9):

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{S^2}\right), \quad (10)$$

where $S > 0$ is the width of the Gaussian kernel that controls the complexity of the SVDD boundary. Given a testing data point $\mathbf{z}$, D^2 that measures the distance between $\mathbf{z}$ and the center, $\mathbf{a}$ can be calculated by the following equation:

$$D^2 = K(\mathbf{z}, \mathbf{z}) - 2 \sum_i \alpha_i K(\mathbf{z}, \mathbf{x}_i) + \sum_{ij} \alpha_i \alpha_j K(\mathbf{x}_i, \mathbf{x}_j). \quad (11)$$

For classification, a new observation $\mathbf{z}$ is classified as the target when D^2 is less than or equal to R^2 .

To illustrate the control boundaries of SVDD, we generated a banana-shaped data set using a Matlab code available from PRTools (Duin *et al.*, 2007). The control boundaries with different values of parameters (f and S) in the SVDD algorithm were constructed from 180 in-control training observations (i.e., Phase I data). Figure 1 shows different SVDD boundaries embedded in two-dimensional plots of Phase I data. It can be seen from Figs. 1(a), (b) and (c) that given the same f value ($f = 0.01$), the shape of the control boundary becomes smoother with larger S . One can choose an appropriate S that balances a trade-off between oversmoothness and undersmoothness of the control boundary. In the present study, we tried some potential values of S and found the one that yields the smallest Type I and Type II error rates. Given the same S value ($S = 3$), Figs. 1 (c), (d), (e) and (f) show that the control boundary becomes tighter to the volume centroid with the larger f .

2.2. Existing control chart methods based on the SVDD algorithm

Several studies have implemented one-class classification methods in SPC problems. Sun and Tsung (2003) proposed K charts to handle non-normality problems by using the kernel distances obtained from the SVDD algorithm. They proposed to establish and adjust the control limits of the K chart by using f (or C), one of the parameters of the SVDD algorithm. Kumar *et al.* (2006) proposed robust K charts, which are similar to K charts but use normalized kernel distances. One-class SVM-based control charts were applied for anomaly detection in computer networks (Zhang *et al.*, 2007). Although the aforementioned control charts use slightly different monitoring statistics, they are all based on the one-class SVM method.

One-Class SVM (OC-SVM)-based control charts can be constructed by plotting monitoring statistics (D^2) that measure the distance between new observations and the center of the hypersphere. The control limits (R^2) of OC-SVM charts are determined by f (or C). In other words, error rates in OC-SVM charts are adjusted by f . Large f values tend to yield a larger Type I error rate because the algorithm utilizes less training data inside the boundary.

Figure 2 displays a T^2 chart and two OC-SVM charts corresponding to the control boundaries in Figs. 1 (b) and (c). In these figures, the monitoring statistics of 400 Phase II data were plotted (the first 360 are in control and the last 40 are out of control). Note that the control limits of these charts were established by 180 Phase I data. In the OC-SVM charts, it is interesting to observe that the user-specified f value affects not only the determination of the control limits but also the calculation of the monitoring statistic. Note that two totally different control charts were obtained by changing the value of f from 0.01 to 0.20 (Figs. 2(c) and (d)). This clearly demonstrates that f is inappropriate for establishing the control limits in OC-SVM charts. This limitation can be explained by Fig. 1, which shows that completely different control boundaries were obtained by changing the value of f from 0.01 to 0.20. As a consequence, an observation detected as being out of control (or in control) may no longer be detected as being out of control (or in control) as a reaction to the use of different values of f . In contrast, T^2 charts use the controlling value α that is independent of the monitoring statistic, T^2 . Thus, the ellipse boundary of T^2 always captures more out-of-control situations and yields a higher Type I error rate with a larger α (Figs. 1(b) and (c)). Furthermore, the same values of monitoring statistics are plotted in the T^2 chart regardless of α (Fig. 2(a)).

2.3. New design strategy of OC-SVM charts based on the bootstrap method

To address the limitation of the current OC-SVM control charts, we propose a new design strategy to establish the

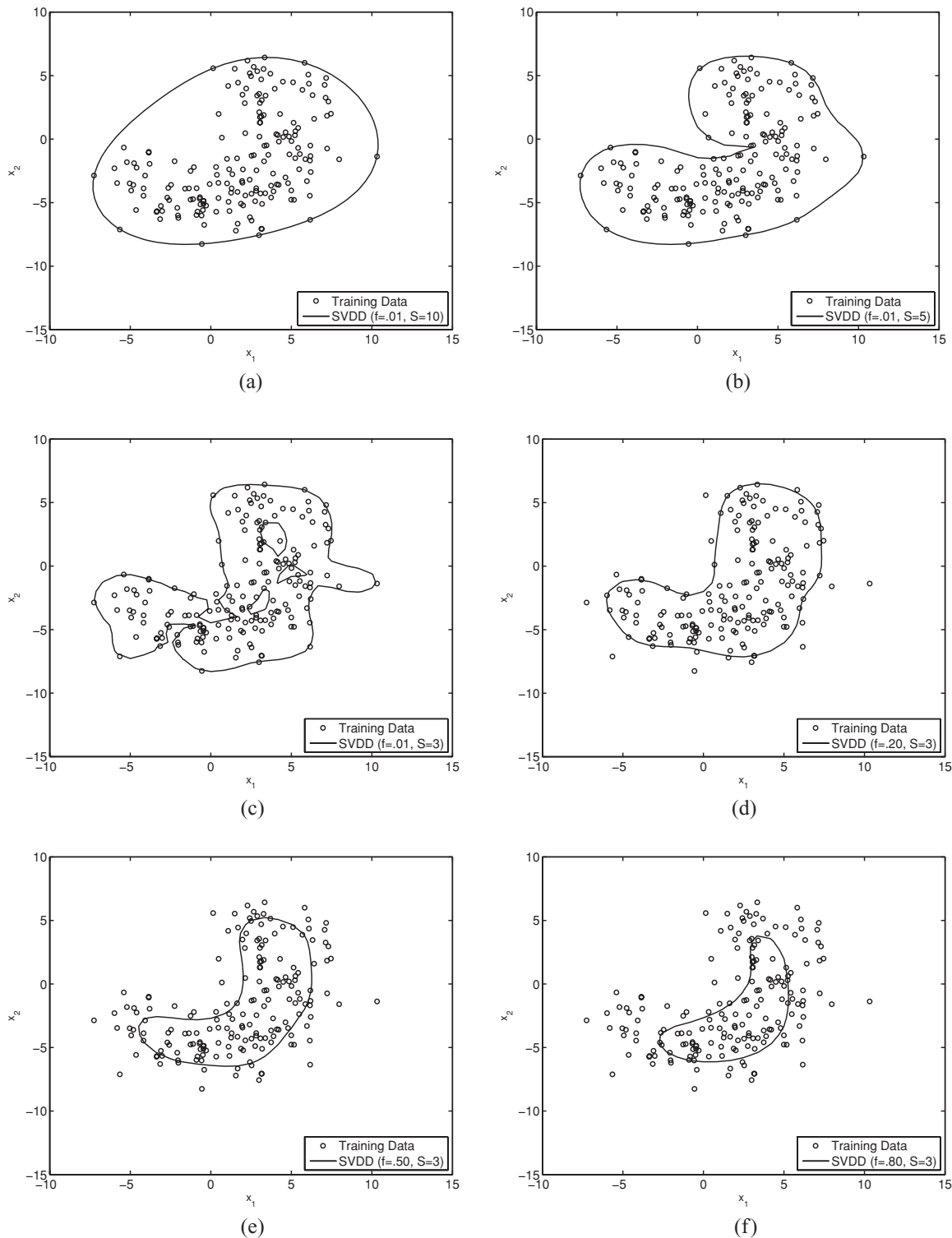


Fig. 1. Control boundaries of SVDD obtained from different values of parameters: (a) $f = 0.01$ and $s = 10$; (b) $f = 0.01$ and $s = 5$; (c) $f = 0.01$ and $s = 3$; (d) $f = 0.20$ and $s = 3$; (e) $f = 0.50$ and $s = 3$; and (f) $f = 0.80$ and $s = 3$.

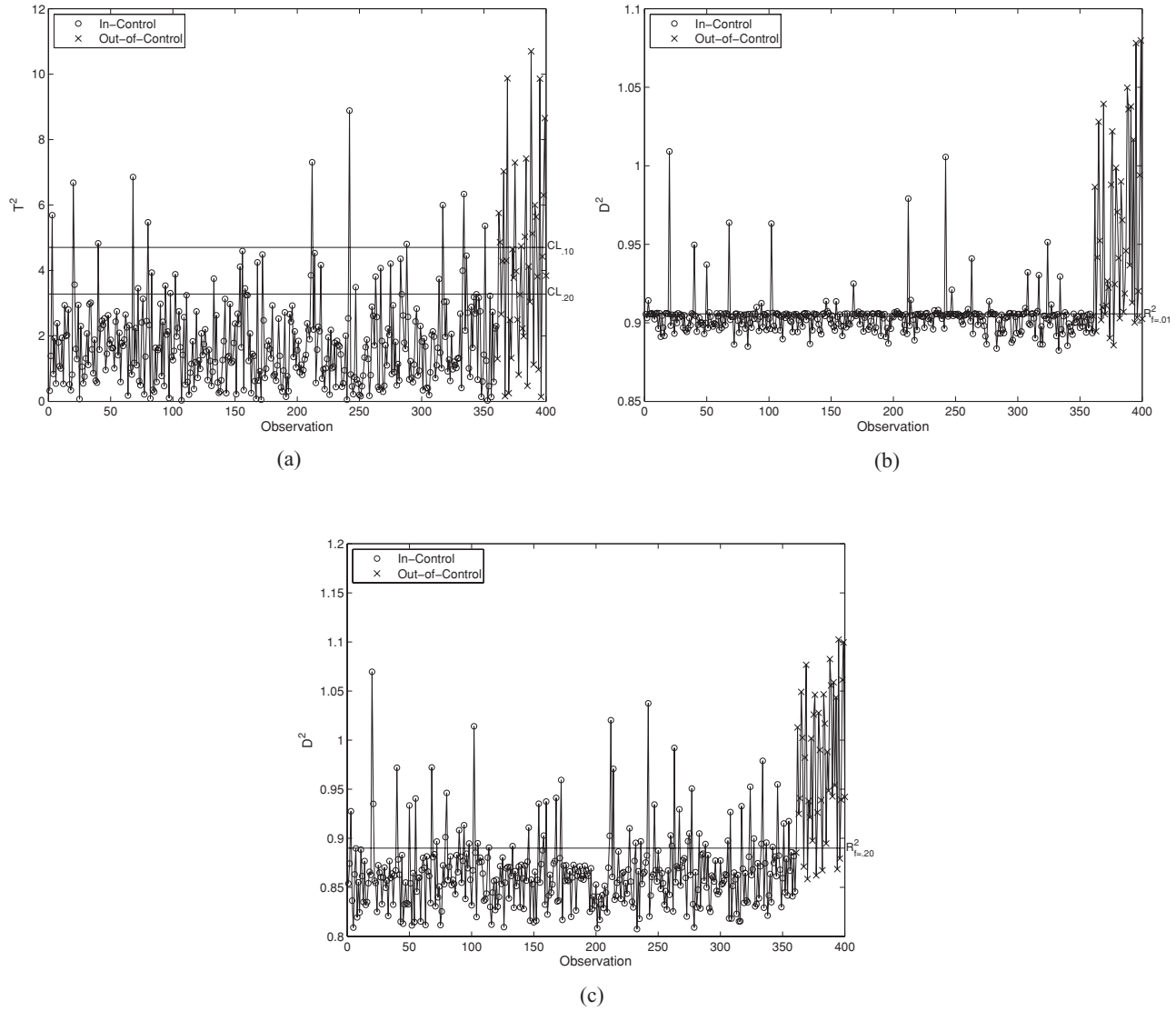


Fig. 2. T^2 and OC-SVM charts with the statistics and the control limits corresponding to the control boundaries in Figs. 1(c) and (d): (a) T^2 chart; (b) OC-SVM chart ($f = 0.01$, $s = 3$); and (c) OC-SVM chart ($f = 0.2$, $s = 3$).

control limits in OC-SVM charts. We call the proposed chart D^2 charts. The control limits of D^2 charts are established and adjusted based on a percentile value estimated by the bootstrap method. The bootstrap method is a resampling method that is widely used to provide statistical estimates when the population distribution is unknown (Efron and Tibshirani, 1993).

In traditional control charts, the control limits are determined based on the underlying distribution of the monitoring statistic with the user-specified value (e.g., Type I error rate). In contrast, the distribution of the monitoring statistic of a D^2 chart is unknown due to its non-parametric nature. This motivates us to develop an appropriate non-parametric procedure to establish the control limit. First,

D^2 values (monitoring statistics) of Phase I observations of size N are obtained through the SVDD algorithm. Second, we take B bootstrap samplings and compute the percentile values of interest from each bootstrap sample of size N drawn with replacement from D^2 values of Phase I observations. Finally, the control limit is determined by taking the average of B percentile values.

The following is a more explicit description of the bootstrap procedure to establish the control limits of the D^2 chart.

Step 1. Compute the D^2 statistics of Phase I observations of size N using Equation (11) and generate B independent bootstrap samples. Let $D_{j1}^2, D_{j2}^2, \dots, D_{jN}^2$

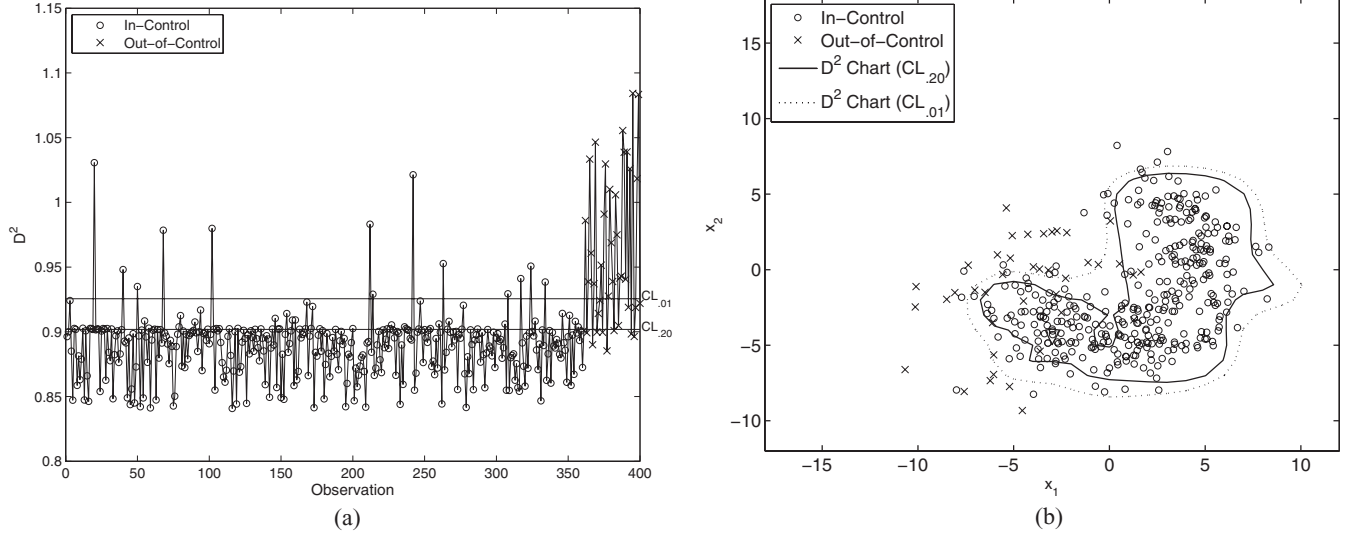


Fig. 3. (a) The D^2 chart; and (b) the corresponding control boundary.

be a sequence of N D^2 statistics from the j th bootstrap sample (for $j = 1, \dots, B$).

Step 2. For each bootstrap sample, given a user-specified α ($0 < \alpha \leq 1$) and the ordered D^2 values ($D_{j(1)}^2 < D_{j(2)}^2 < \dots < D_{j(N)}^2$), $D_{j(i)}^2$ is the i th largest value of N D^2 values in the j th bootstrap sample where i is a roundup number of $N \times \alpha$.

Step 3. Calculate the control limit (CL) by taking average of the i th largest values (i.e., $100(1 - \alpha)$ th percentile values) in each of B bootstrap samples: $CL = \sum_{j=1}^B D_{j(i)}^2 / B$.

Step 4. Monitoring Phase II observations: declare the observations out of control if the corresponding D^2 values exceed the control limit.

Figure 3 displays the D^2 chart and the corresponding control boundary. In the D^2 control chart, 180 in-control observations were used to estimate the control limits (bootstrap-based 99th and 80th percentiles of the D^2 statistics) and 400 D^2 statistics from Phase II observations were plotted. Figure 3(b) shows the corresponding control boundary generated from the D^2 chart in Fig. 3(a). It can be seen that by increasing the α value from 0.01 to 0.20, more out-of-control observations were detected.

3. k -nearest neighbors data description-based control charts

The SVDD algorithm involves an optimization problem that requires a high computational load during the training process. The SVDD algorithm requires around 4.06 hours in one of our machines to train the model using 4000 bivariate observations. Because of the high computational

cost, D^2 charts may not be efficient for a process that needs frequent retraining. In order to address this computational burden, we propose a new one-class classification-based control chart called a K^2 chart. The algorithm used in a K^2 chart requires about 5.42 seconds (on the same machine as the SVDD algorithm) to complete 4000 bivariate training observations. K^2 charts are based on a k -Nearest Neighbors Data Description (k NNDD) method that solves one-class classification problems by estimating the local density of the data using a nearest neighbors algorithm (Breunig *et al.*, 2000; Tax, 2001). A brief description of the k NNDD algorithm is presented in the following section.

3.1. The k NNDD algorithm

Let $NN_i(\mathbf{z})$ be the i th nearest-neighbor training observation of a data point $\mathbf{z}$ that needs to be classified (or monitored). Let V be the volume of the hypersphere containing i nearest-neighbor training observations. Let N be the size of the training set. The local density of $\mathbf{z}$ can be determined by

$$d(\mathbf{z}) = \frac{i/N}{V\|\mathbf{z} - NN_i(\mathbf{z})\|}. \quad (12)$$

Similarly, the local density of $NN_i(\mathbf{z})$ can be determined by

$$d(NN_i(\mathbf{z})) = \frac{i/N}{V\|NN_i(\mathbf{z}) - NN_i(NN_i(\mathbf{z}))\|}, \quad (13)$$

where $NN_i(NN_i(\mathbf{z}))$ is the i th nearest neighbor of $NN_i(\mathbf{z})$ in the same training set. The k NNDD algorithm classifies $\mathbf{z}$ as the target class when the ratio of its local density of $\mathbf{z}$ (Equation (12)) to the local density of $NN_i(\mathbf{z})$ (Equation (13)) is greater than or equal to one, which can be

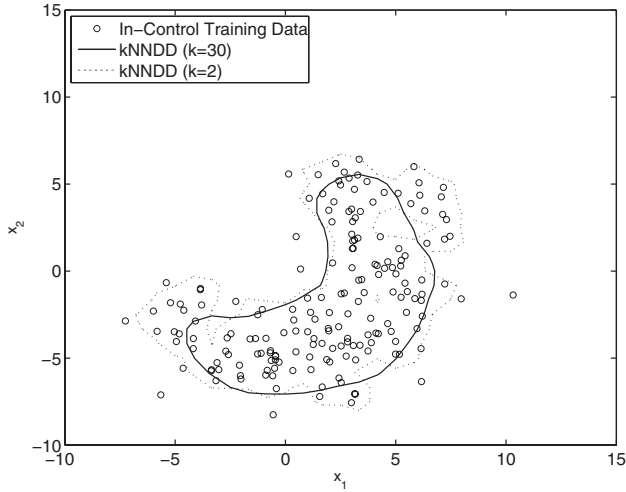


Fig. 4. Control boundaries of k NNDD (with different k) constructed from the banana-shaped data set.

explained as follows:

$$\frac{d(\mathbf{z})}{d(\text{NN}_i(\mathbf{z}))} = \frac{\|\text{NN}_i(\mathbf{z}) - \text{NN}_i(\text{NN}_i(\mathbf{z}))\|}{\|\mathbf{z} - \text{NN}_i(\mathbf{z})\|} \geq 1. \quad (14)$$

To make the algorithm more robust, the average of k distances is considered (for $i = 1, \dots, k$). Thus, Equation (14) becomes:

$$\frac{\sum_{i=1}^k \|\text{NN}_i(\mathbf{z}) - \text{NN}_i(\text{NN}_i(\mathbf{z}))\|}{\sum_{i=1}^k \|\mathbf{z} - \text{NN}_i(\mathbf{z})\|} \geq 1. \quad (15)$$

In the k NNDD algorithm, the size of nearest neighbor, k , affects its performance. Figure 4 displays the control boundaries obtained by k NNDD with two different values

of k . The decision boundary with $k = 30$ is fairly smooth compared to the control boundary obtained by using $k = 2$. One can search possible values of k and find an appropriate one that compromises a trade-off between oversmoothness and undersmoothness of the control boundary. A previous study indicated that the proper range of k in the k NNDD algorithm is between ten and 50 (Breunig *et al.*, 2000).

3.2. K^2 charts

To construct K^2 charts, the average distance between $\mathbf{z}$ and k nearest observations is calculated as follows:

$$K^2 = \frac{\sum_{i=1}^k \|\mathbf{z} - \text{NN}_i(\mathbf{z})\|}{k}. \quad (16)$$

K^2 values are then used as monitoring statistics. The control limits of a K^2 chart are obtained by the bootstrap percentile procedure as we proposed in D^2 charts (please see Section 2.3). Here is the detailed summary of the bootstrap percentile procedure for K^2 charts.

- Step 1.* Compute the D^2 statistics of Phase I observations of size N using Equation (11) and generate B independent bootstrap samples. Let $K_{j1}^2, K_{j2}^2, \dots, K_{jN}^2$ be a sequence of N K^2 statistics from the j th bootstrap sample (for $j = 1, \dots, B$).
- Step 2.* For each bootstrap sample, given a user-specified α ($0 < \alpha \leq 1$) and the ordered K^2 values ($K_{j(1)}^2 < K_{j(2)}^2 < \dots < K_{j(N)}^2$), $K_{j(i)}^2$ is the i th largest value of N K^2 values in the j th bootstrap sample where i is a roundup number of $N \times \alpha$.
- Step 3.* Calculate the control limit (CL) by taking average of the i th largest values (i.e., $100(1 - \alpha)$ th

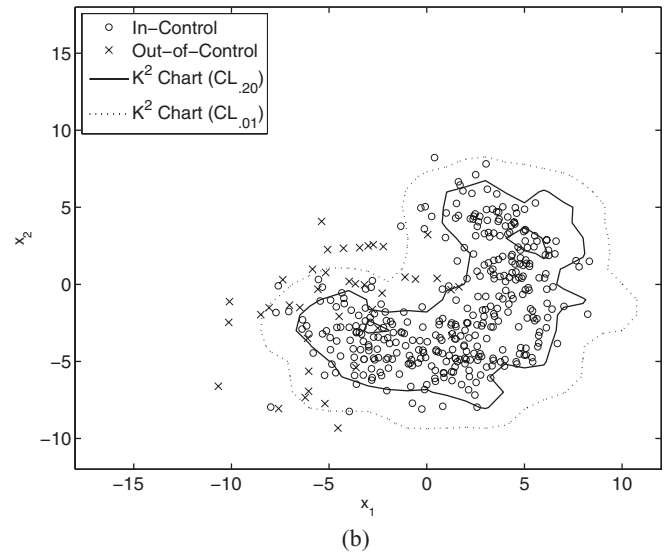
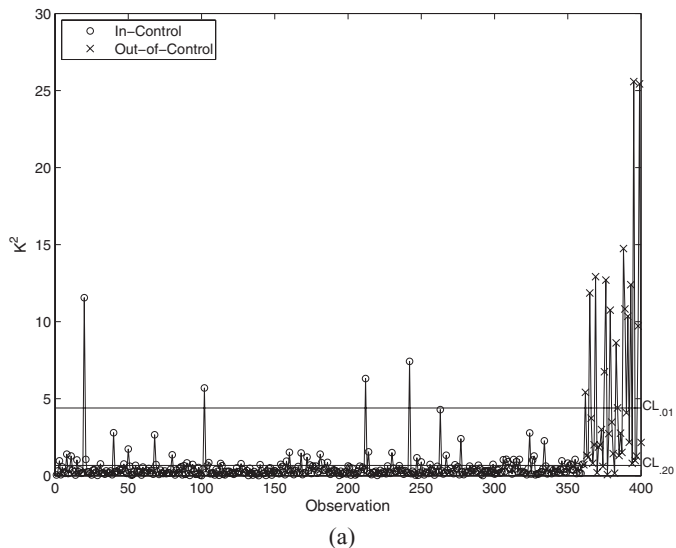


Fig. 5. (a) The K^2 chart; and (b) the corresponding control boundary.

percentile values) in each of B bootstrap samples:
 $CL = \sum_{j=1}^B K_{j(i)}^2 / B.$

Step 4. Monitor Phase II observations: declare the observations out of control if the corresponding K^2 values exceed the control limit.

Figure 5 displays the K^2 chart ($k = 30$) and the corresponding control boundary from the banana-shaped data set. Two different control limits were calculated by estimated percentiles (99th and 80th) from 5000 bootstrap samples of 180 K^2 statistics. For monitoring Phase II observations, the K^2 value of each Phase II observation was plotted. Figure 5(b) displays the control boundaries corresponding to the control limits embedded in a two-dimensional plot

of the Phase II observations, showing that control charts become more sensitive as α increases.

4. Simulation study

4.1. Simulation setup

A simulation study was conducted to compare the performance among D^2 , K^2 , T^2 and OC-SVM charts. We generated the data based on the bivariate normal, bivariate t , bivariate gamma and a banana-shaped data set. For D^2 and OC-SVM charts, we used the width of Gaussian kernel, $S = 1$ for the normal, t , and gamma cases and $S = 3$ for the banana-shaped data. For K^2 charts, we used $k = 30$.

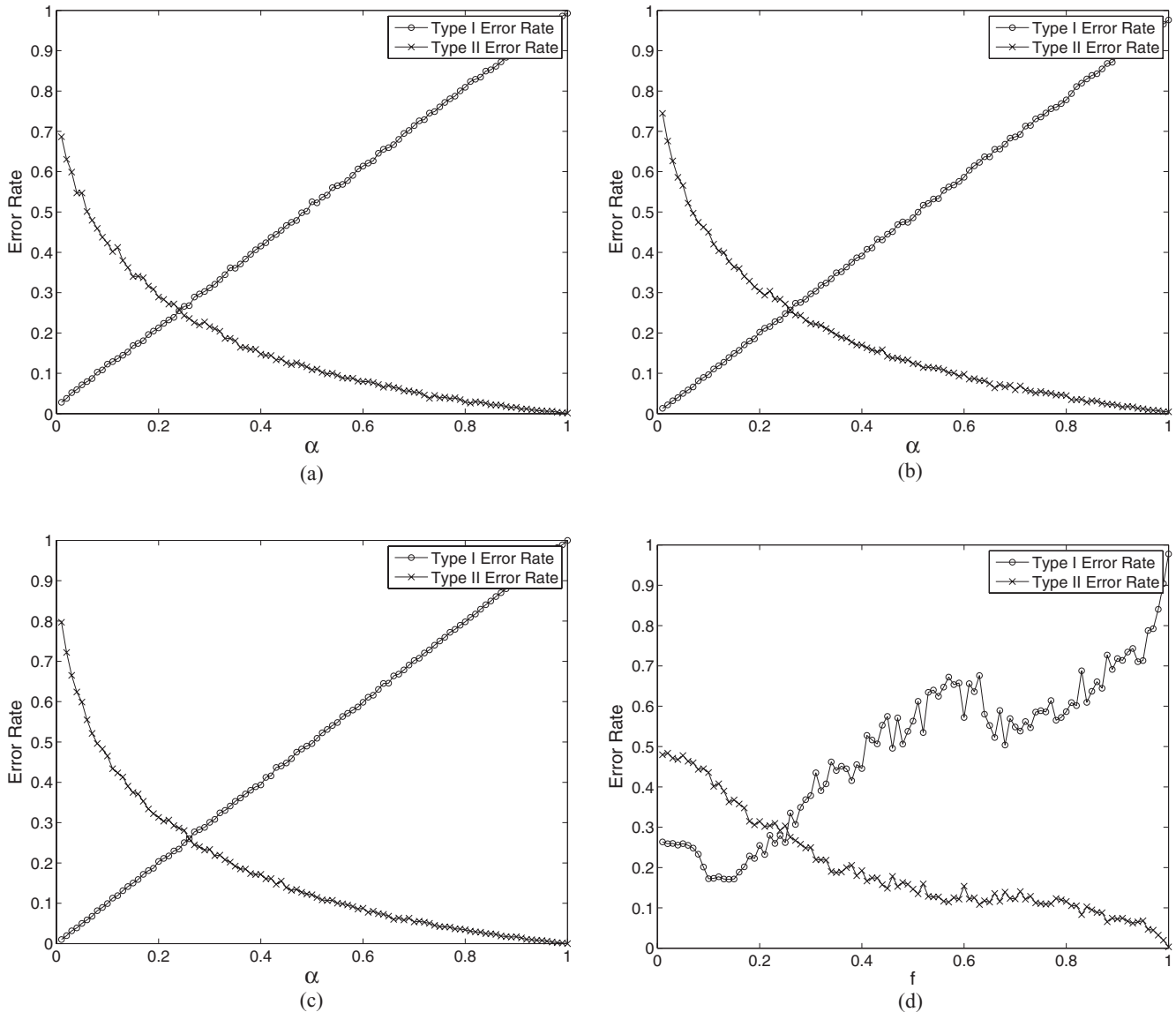


Fig. 6. Average Type I and Type II error rates from: (a) D^2 ; (b) K^2 ; (c) T^2 ; and (d) OC-SVM charts (N_2 with $\lambda = 2$ scenario).

One thousand Phase II observations (900 in control and 100 out of control) were monitored based on the control limits that were established by 200 Phase I observations. Let μ_0 and Σ_0 be the mean vector and the covariance matrix of the in-control data. Let $\mu_1 = \mu_0 + \delta$ be the mean vector of the out-of-control data. The magnitude of the shift δ is represented by the following non-centrality parameter λ

$$\lambda = \sqrt{\delta^T \Sigma_0^{-1} \delta}. \quad (17)$$

To generate the out-of-control data for the bivariate normal, bivariate t and bivariate gamma distributions, two types of mean shifts (i.e., the medium mean shift $\lambda = 2$ and the large mean shift $\lambda = 3$) were considered. At a certain value of λ , all variables are shifted equally. Note that

we do not consider the change in variance. We generated two different angles of banana shapes that represent the in-control and out-of-control data (please see Duin *et al.* (2007) for more details on generating the banana-shaped data set). The summary of simulation scenarios is described as follows.

1. $N_2, \lambda = 2$: The medium-mean-shift case of the bivariate normal distribution with $\mu_0 = [0 \ 0]$ and $\Sigma_0 = \begin{bmatrix} 1 & 0.35 \\ 0.35 & 1 \end{bmatrix}$.
2. $N_2, \lambda = 3$: The large-mean-shift case of the bivariate normal distribution with $\mu_0 = [0 \ 0]$ and $\Sigma_0 = \begin{bmatrix} 1 & 0.35 \\ 0.35 & 1 \end{bmatrix}$.

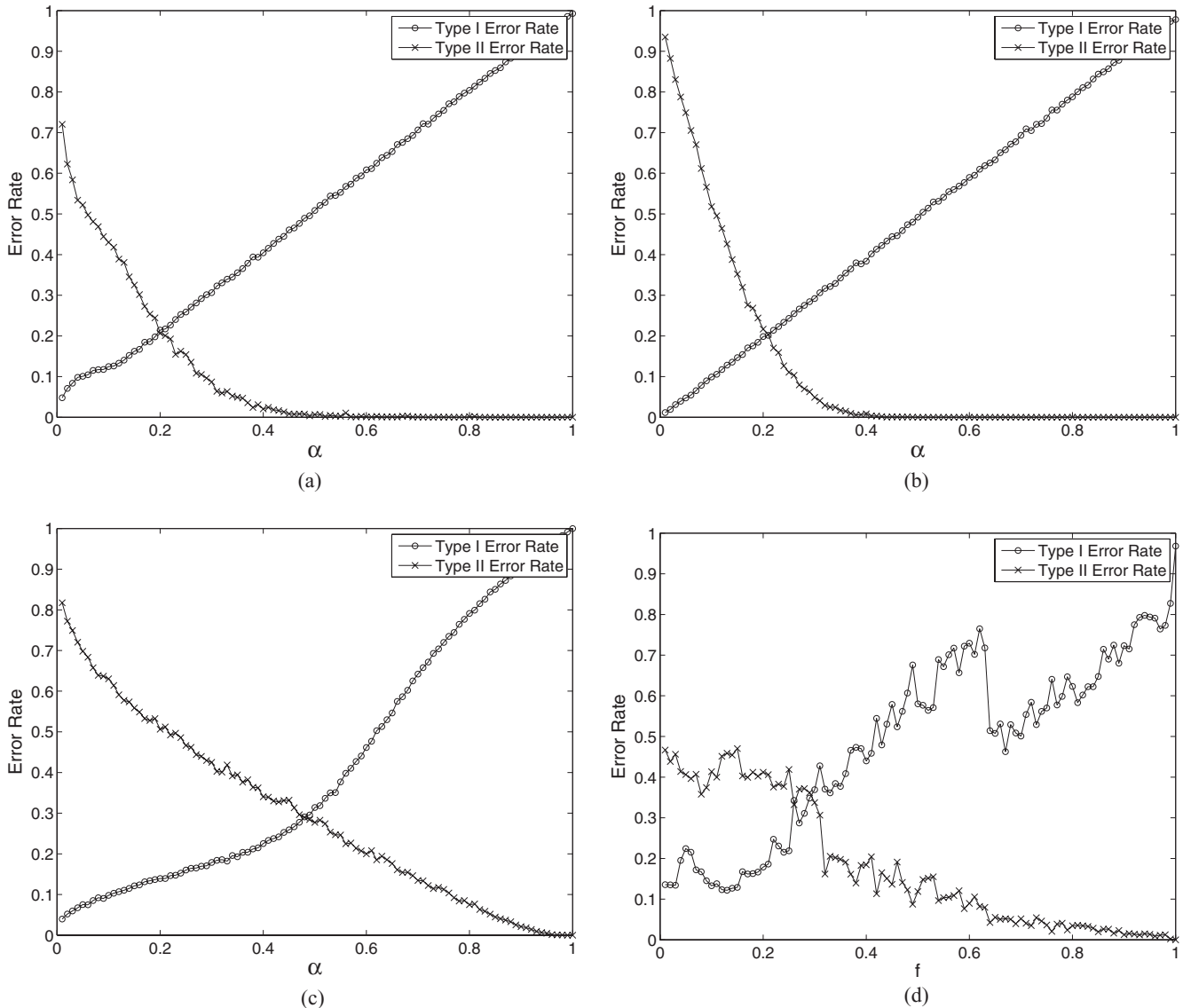


Fig. 7. Average Type I and Type II error rates from: (a) D^2 ; (b) K^2 ; (c) T^2 ; and (d) OC-SVM charts ($Gam_2(1, 1)$ with $\lambda = 2$ scenario).

3. $t_2(3)$, $\lambda = 2$: The medium-mean-shift case of the bivariate t distribution with three degrees of freedom.
4. $t_2(3)$, $\lambda = 3$: The large-mean-shift case of the bivariate t distribution with three degrees of freedom.
5. $Gam_2(1, 1)$, $\lambda = 2$: The medium-mean-shift case of the bivariate gamma distribution with the shape and scale parameters, where both of them are one.
6. $Gam_2(1, 1)$, $\lambda = 3$: The large-mean-shift case of the bivariate gamma distribution with the shape and scale parameters, where both of them are one.
7. Banana-shaped: A banana-shaped data set with two different angles.

4.2. Control limits

In contrast to existing OC-SVM charts that use the parameter f of the SVDD algorithm to adjust the control limits, the control limits of D^2 and K^2 charts are adjusted by the percentile, which is estimated by the bootstrap method. Figures 6 and 7 show how actual Type I and Type II error rates in the D^2 , K^2 , T^2 and OC-SVM charts are controlled by the controlling factors (α or f), indicated in the x -axes. We used the average values of actual Type I and Type II error rates from 100 simulation runs. The standard errors of 100 simulations are relatively small (between 0.02 and 0.06), demonstrating that 100 simulations are enough to draw the meaningful conclusion. We presented the results for only N_2 and $Gam_2(1, 1)$ scenarios, respectively, as examples of normal and non-normal cases. In general, as the controlling factor increases, all control charts produced larger Type I error rates but produced smaller Type II error rates. The particularly strong positive correlation between

the actual Type I error rate and the controlling factor is desired. The proposed D^2 and K^2 charts satisfy this condition in both normal and non-normal cases, but T^2 charts satisfy this condition in only normal cases. In both normal and non-normal cases, OC-SVM charts failed to provide strong linear correlation between the actual Type I error rate and the controlling factor. Moreover, Type I and Type II error rates may not be properly controlled by f as the size of target observations goes up in OC-SVM charts. Figure 8 shows OC-SVM charts constructed from the N_2 with $\lambda = 2$ scenario using 300 and 400 target observations. It can be observed that Type I and Type II error rates seem to be constant over the different values of f . As we defined earlier in Equation (4), f represents the fraction of the target data outside the decision boundary and has an inverse relationship with the total number of target observations. Thus, with the large number of target observations, the fraction of the target data (f) only plays a small role in changing the control boundary, leading to relatively constant Type I and Type II error rates. These demonstrate that f is an inappropriate choice as the controlling factor in OC-SVM charts.

4.3. Performance comparisons

The average values of Type I and Type II error rates from 100 simulation runs among D^2 , K^2 , T^2 and OC-SVM charts were compared. The control chart that yields a lower Type II error rate is considered a better method if the Type I error rate is similar. Figure 9 displays the average rates of Type I and Type II error under all the simulation scenarios studied.

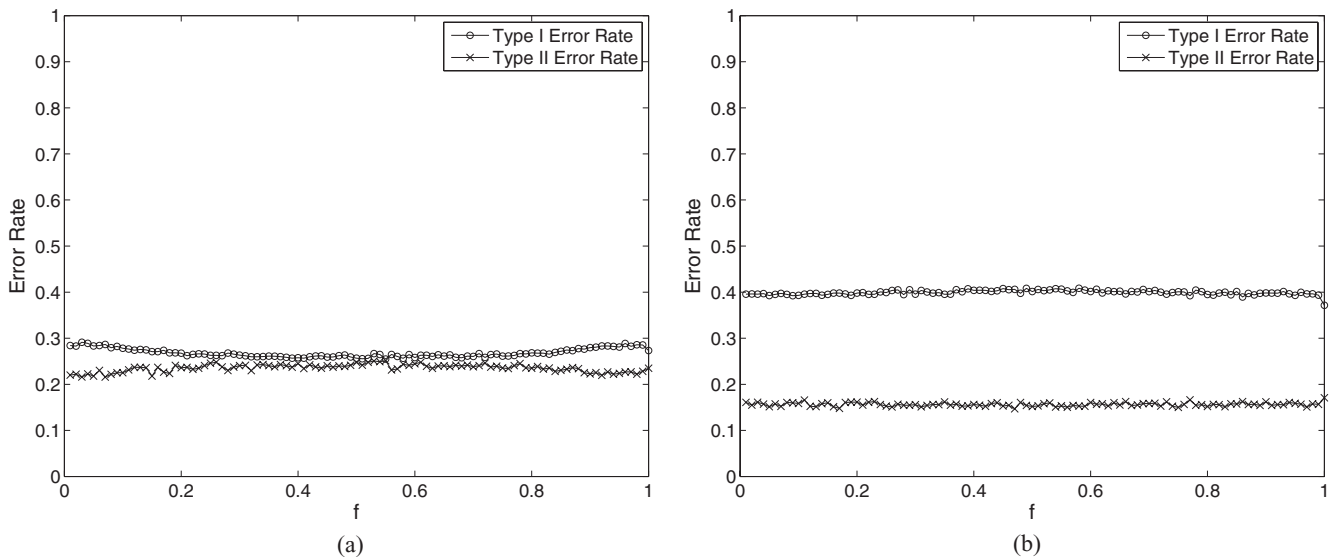


Fig. 8. Average Type I and Type II error rates from OC-SVM charts when the number of Phase I observations is large (N_2 with $\lambda = 2$ scenario): (a) 300 observations; and (b) 400 observations.

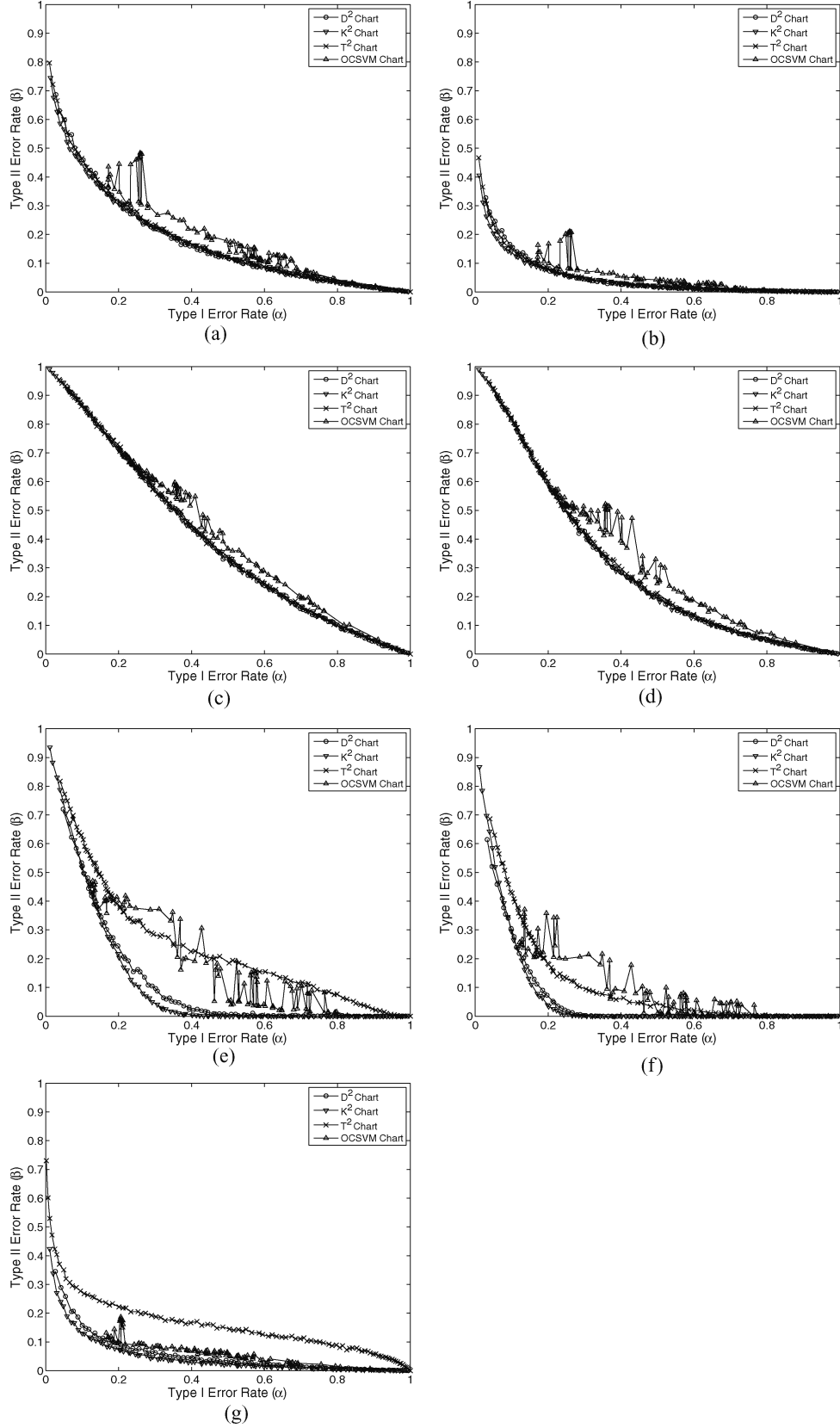


Fig. 9. Type I and Type II error rates of the D^2 , K^2 , T^2 and OC-SVM charts for the studied simulation scenarios: (a) N_2 with $\lambda = 2$ scenario; (b) N_2 with $\lambda = 3$ scenario; (c) $t_2(3)$ with $\lambda = 2$ scenario; (d) $t_2(3)$ with $\lambda = 3$ scenario; (e) $Gam_2(1, 1)$ with $\lambda = 2$ scenario; (f) $Gam_2(1, 1)$ with $\lambda = 3$ scenario; and (g) banana-shaped scenario.

Table 1. Average values of Type I error rate (α) and Type II error rate (β) of the D^2 chart the K^2 chart and the recursive T^2 in Phase I application (average values of standard errors are shown inside the parentheses)

Scenarios	D^2		K^2		T^2	
	α	β	α	β	α	β
$N_2, \lambda = 2$	0.1869 (0.0128)	0.3630 (0.0994)	0.1829 (0.0124)	0.3485 (0.1065)	0.1892 (0.0566)	0.3865 (0.1354)
$N_2, \lambda = 3$	0.2124 (0.0103)	0.0800 (0.0674)	0.2183 (0.0097)	0.0825 (0.0561)	0.2137 (0.0664)	0.0915 (0.0810)
$t_2(3), \lambda = 2$	0.2219 (0.0119)	0.6830 (0.0932)	0.2144 (0.0122)	0.6790 (0.0970)	0.2248 (0.0426)	0.7165 (0.1071)
$t_2(3), \lambda = 3$	0.2038 (0.0132)	0.6290 (0.0949)	0.1995 (0.0126)	0.6375 (0.1013)	0.2069 (0.0444)	0.6230 (0.1436)
$Gam_2(1), \lambda = 2$	0.1996 (0.0171)	0.2410 (0.1307)	0.2101 (0.0140)	0.2825 (0.1196)	0.2089 (0.0596)	0.3250 (0.1969)
$Gam_2(1), \lambda = 3$	0.2204 (0.0100)	0.0380 (0.0556)	0.2208 (0.0116)	0.0715 (0.0905)	0.2335 (0.0781)	0.1200 (0.1482)
Banana-shaped	0.1751 (0.0118)	0.1680 (0.0886)	0.1771 (0.0127)	0.0865 (0.0721)	0.1791 (0.1026)	0.2380 (0.1211)

The result shows that the D^2 and K^2 charts produced smaller Type II error rates than the T^2 chart, given similar Type I error rates in the gamma and banana-shaped data scenarios. In the normal and t cases, all methods provide comparable performances. The range of standard errors of 100 simulation is between 0.02 and 0.08 for the normal, t and gamma cases, while much larger standard errors were obtained from OC-SVM (between 0.10 and 0.26). It should be noted that OC-SVM charts produce irregular Type II error rates over Type I error rates. Consequently, it is difficult to compare the performance of OC-SVM with other charts.

5. Phase I application of D^2 and K^2 charts

Phase I analysis separates the in-control data from the historical data set, which is a mixture of in-control and out-of-control data, in order to establish the reliable control limits for monitoring future observations. A simulation study was conducted to show the applicability of D^2 and K^2 charts for Phase I problems. We compared the performance of the D^2 and K^2 charts with the existing Phase I method that recursively removes the observations that exceed the control limits until no out-of-control observations are detected. In multivariate processes, this recursive procedure is performed by Hotelling's T^2 control chart in the Phase I application (Montgomery, 2005).

We generated 200 historical observations from the bivariate normal, bivariate t , and bivariate gamma distributions and a banana-shaped data set. We assigned 20 observations (out of 200) to be out of control where two different non-centrality parameters, $\lambda = 2$ and $\lambda = 3$, were used for the bivariate normal, bivariate t and bivariate gamma distributions. For the banana-shaped data set, two different

angles of banana shapes were used to represent the in-control and out-of-control data. The D^2 and K^2 charts were constructed with all 200 observations. The control charts removed the historical observations in which the statistics exceed the control limits. Analogous to D^2 and K^2 charts for Phase II analysis, $100(1 - \alpha)$ th bootstrap percentiles of the D^2 and K^2 statistics of the historical data were used as control limits in Phase I analysis. The remaining observations were defined as being in control. The observations that were actually in control but incorrectly removed were Type I errors. The remaining observations that were actually out of control were Type II errors.

We compared the performances of the D^2 and K^2 charts with the recursive T^2 in terms of Type I and Type II error rates (average values from 100 simulation runs). Table 1 shows that the performances of the D^2 and K^2 charts are slightly better than recursive T^2 under the normal and t scenarios but they are comparable. Because Hotelling's T^2 chart can effectively handle multivariate normal data, the recursive T^2 is also an appropriate method in normal distribution cases of Phase I analysis. However, in the gamma and banana-shaped data scenarios, the D^2 and K^2 charts produced smaller Type II error rates than the recursive T^2 method. This clearly demonstrates that D^2 and K^2 charts are effective approaches to use for Phase I analysis in both normal and non-normal cases.

6. Conclusions

We have proposed new multivariate control charts based on one-class classification algorithms. The proposed D^2 and K^2 charts obtain their monitoring statistics from the SVDD and k NNDD algorithms. The control limits

are derived from the bootstrap-estimated percentile of monitoring statistics. The proposed control charts, because of their data-driven nature, can effectively describe reality, reflect the unique characteristics of the data being monitored, and require a minimal set of assumptions to construct a control chart. The comparative study from the simulated data shows that performances of the D^2 and K^2 charts were comparable to T^2 charts in the normal distribution case. However, D^2 and K^2 charts outperformed T^2 charts in non-normal distribution cases. Moreover, we demonstrated the applicability and effectiveness of the D^2 and K^2 chart techniques for Phase I problems. There are several interesting directions for future research. One such direction is to extend our study to other one-class classification methodologies. Another research directions to develop more efficient ways to establish control limits. A more comprehensive simulation study should be conducted to evaluate the efficacy and consequences of various scenarios, including the impact of variance changes.

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Biographies

Thuntee Sukchotrat is a Faculty Associate Researcher in the Industrial and Manufacturing Systems Engineering Department at the University of Texas at Arlington (UTA). He received a B.E. in Industrial Engineering from Chulalongkorn University, Thailand, and an M.S. and a Ph.D. in Industrial Engineering from UTA. His main research interests include data mining and statistical process control.

Seoung Bum Kim is an Assistant Professor in the Department of Information Management Engineering at Korea University. From 2005 to 2009, he was a faculty member of the Industrial and Manufacturing Systems Engineering Department at the University of Texas at Arlington. He received an M.S. in Industrial and Systems Engineering in 2001, an M.S. in Statistics in 2004 and a Ph.D. in Industrial and Systems Engineering at the Georgia Institute of Technology. He was awarded the Jack

Youden Prize as the best expository paper in *Technometrics* for the Year 2003. He is a member of the Institute for Industrial Engineers, Institute for Operations Research and Management Science and the Institute of Mathematical Statistics. He is actively involved in the Data Mining cluster and Quality, Reliability, and Statistics cluster by presenting papers and chairing sessions at the INFORMS annual and international meetings. He has also served as a council member, a newsletter editor, and a web editor of the INOFRMS Section on Data Mining. His research interests are theory and applications of data mining algorithms and statistical quality control.

Fugee Tsung is Professor and Head of the Department of Industrial Engineering and Logistics Management (IELM), Director of the Quality Laboratory, at the Hong Kong University of Science & Technology

(HKUST). He received both his M.Sc. and Ph.D. from the University of Michigan, Ann Arbor. He is currently Associate Editor of *Technometrics*, Department Editor of the *IIE Transactions*, and on the Editorial Boards for *Quality and Reliability Engineering International*, the *International Journal of Reliability, Quality and Safety Engineering*, among others. He is an ASQ Certified Six Sigma Black Belt, ASQ authorized Six Sigma Master Black Belt Trainer, Co-founder and Chair of the Service Science Section at the Institute for Operations Research and the Management Sciences (INFORMS), and Regional Vice President (Asia) of the Institute of Industrial Engineers. He has authored over 70 refereed journal publications, and is also the winner of the Best Paper Award for the *IIE Transactions* in 2003. His research interests include quality engineering and management to manufacturing and service industries, statistical process control, monitoring and diagnosis.